

Ethanol Blending in Gasoline – Finland

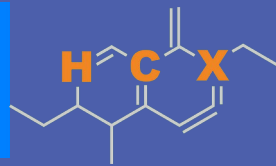
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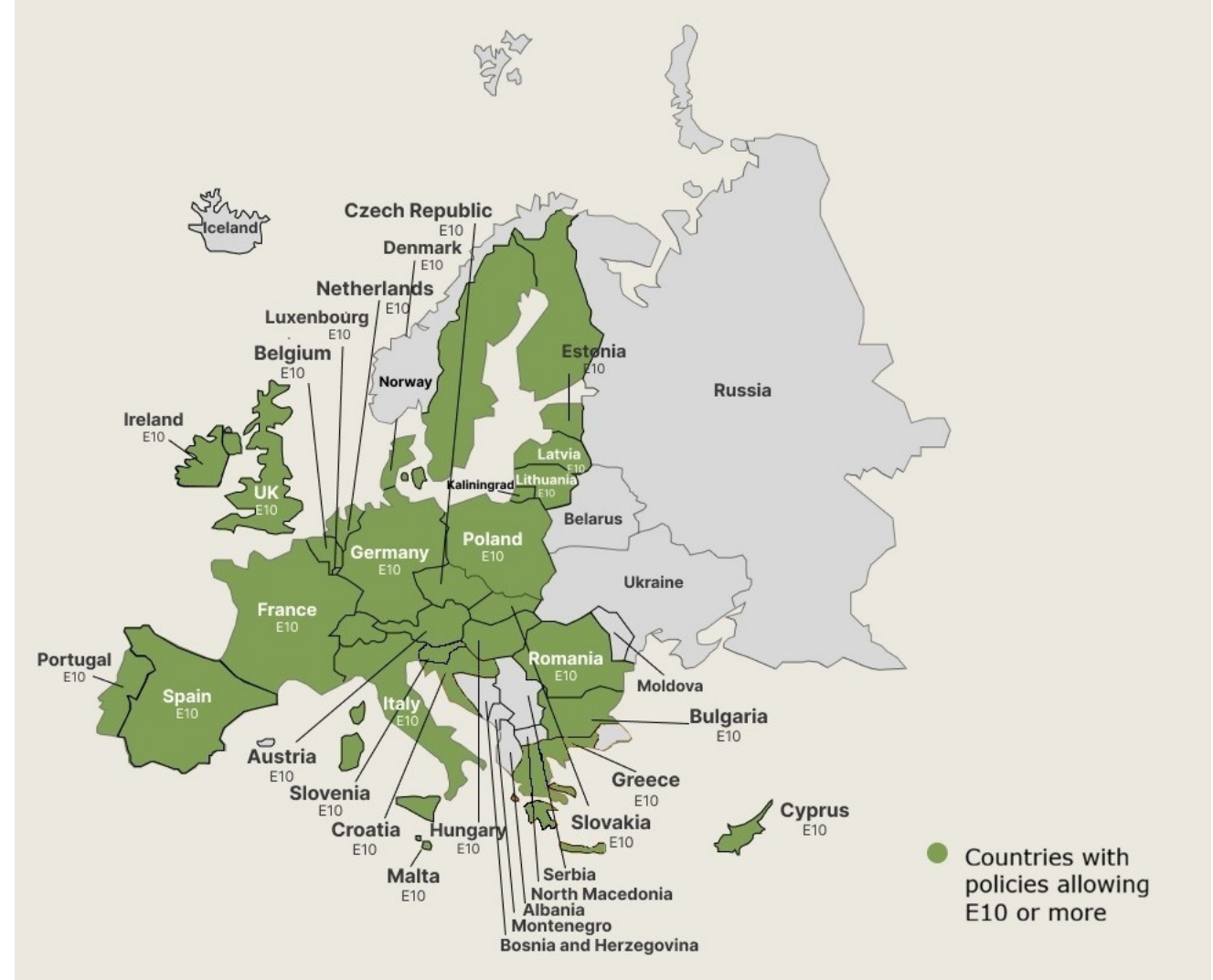


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

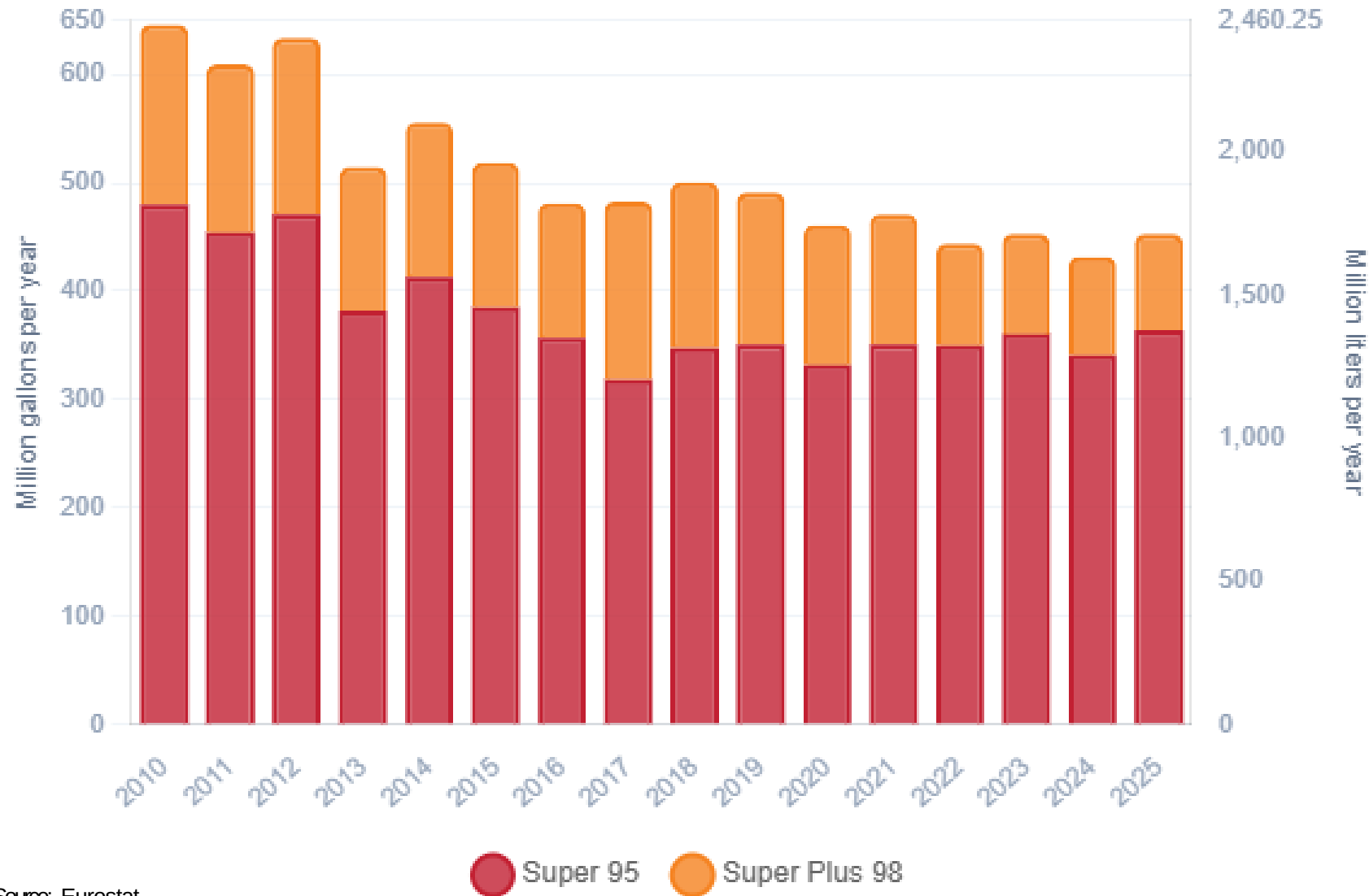
Ethanol Gasoline Blending in Finland



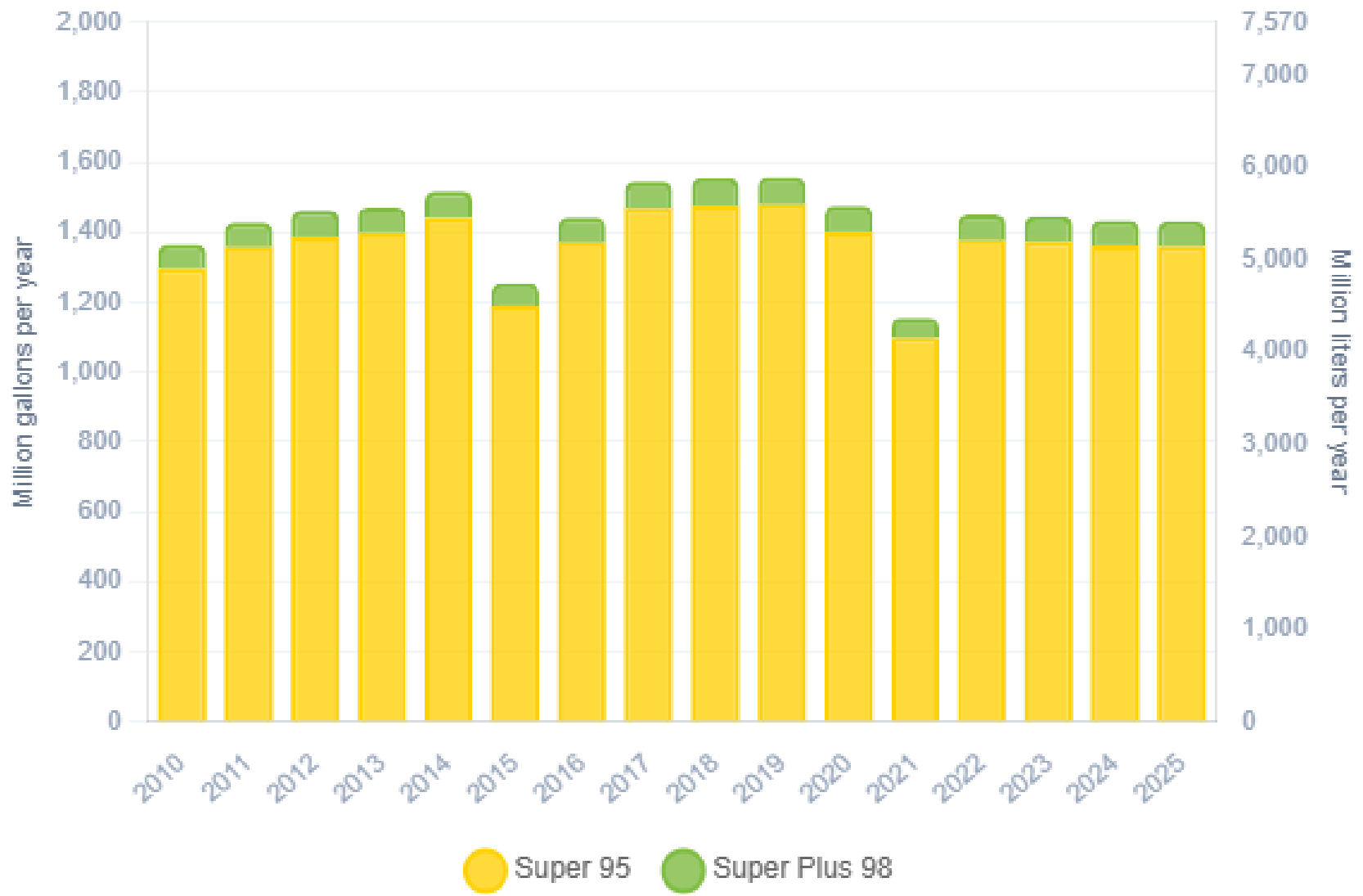
In 2025, gasoline consumption in Finland was 1,723 million liters (455 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 5,696 and -3,919 million liters (1,505 and -1,035 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

In 2025, ethanol consumption in Finland was 155 million liters (41 million gallons) representing 9% of gasoline demand. Ethanol production and Net Imports were 23 and 87 million liters (6 and 23 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

Gasoline Demand in Finland

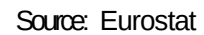


Gasoline Production in Finland

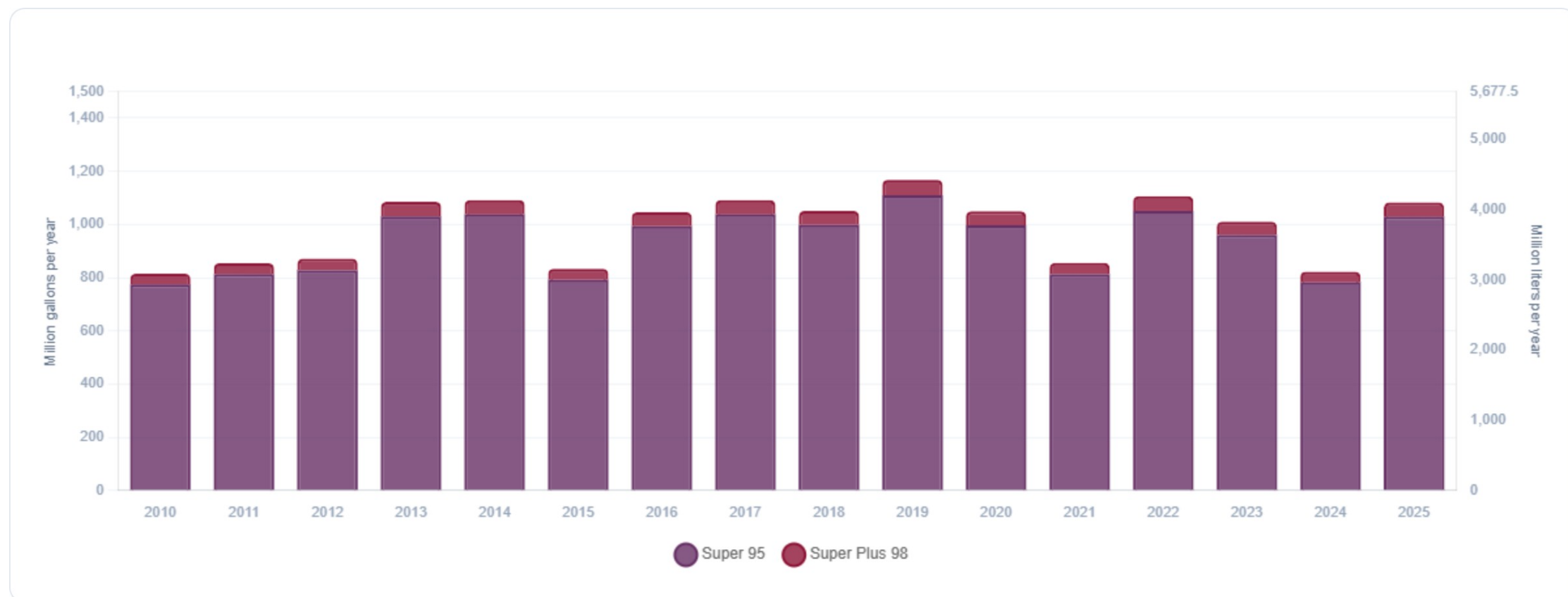


Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups and a propyl group attached.

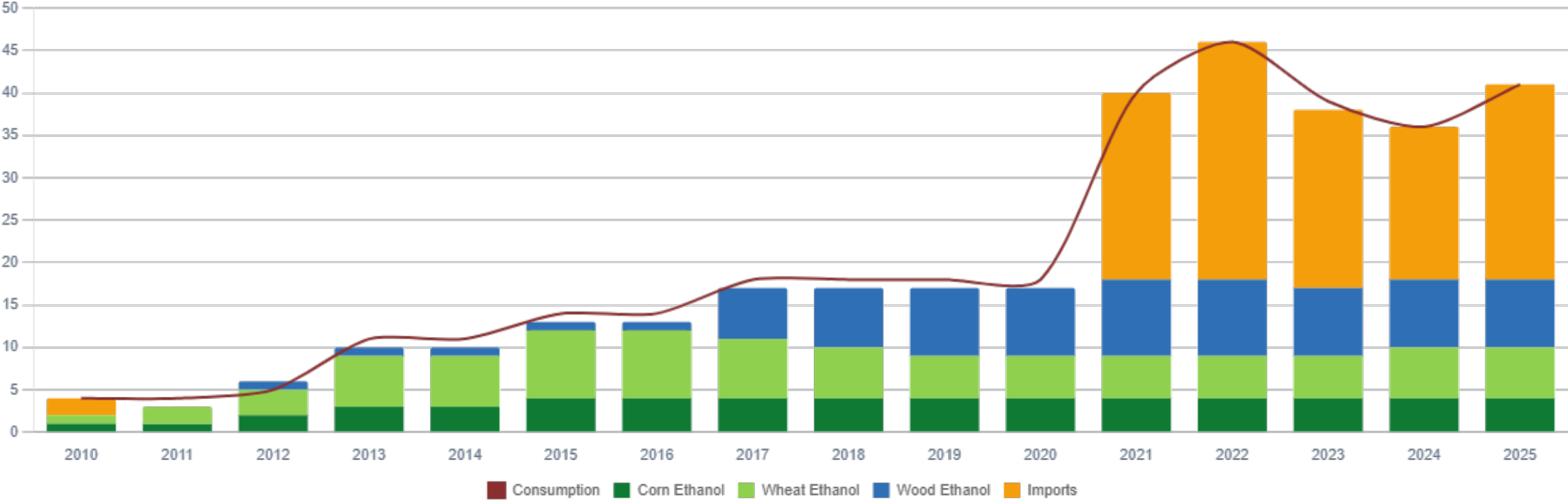


Gasoline Exports in Finland



Source: Eurostat

Ethanol Balance in Finland



Source: Eurostat

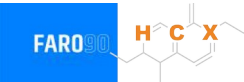
Gasoline Quality in Finland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	< > 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	16.5							
Advanced Biofuels (%cal)	4.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

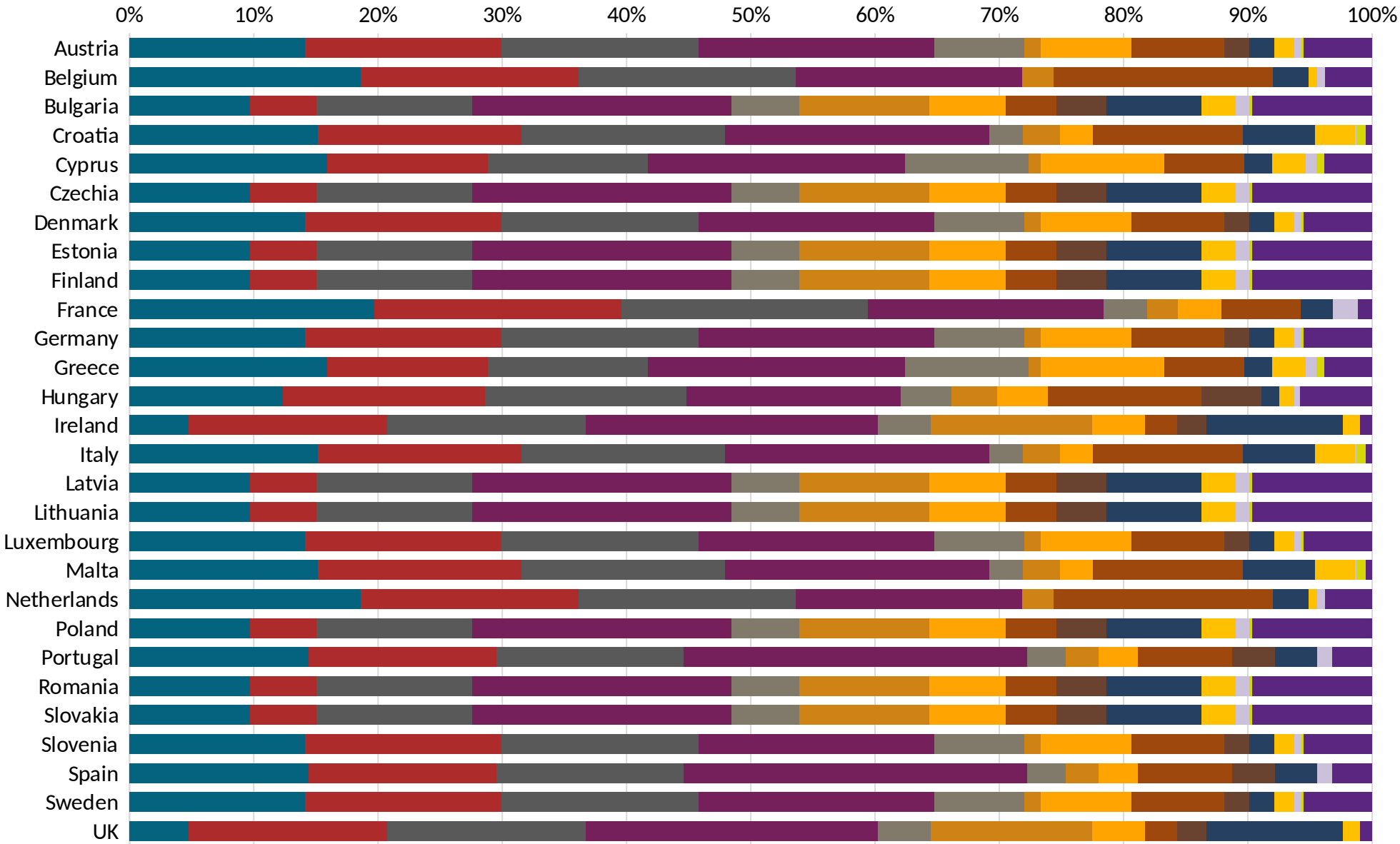
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



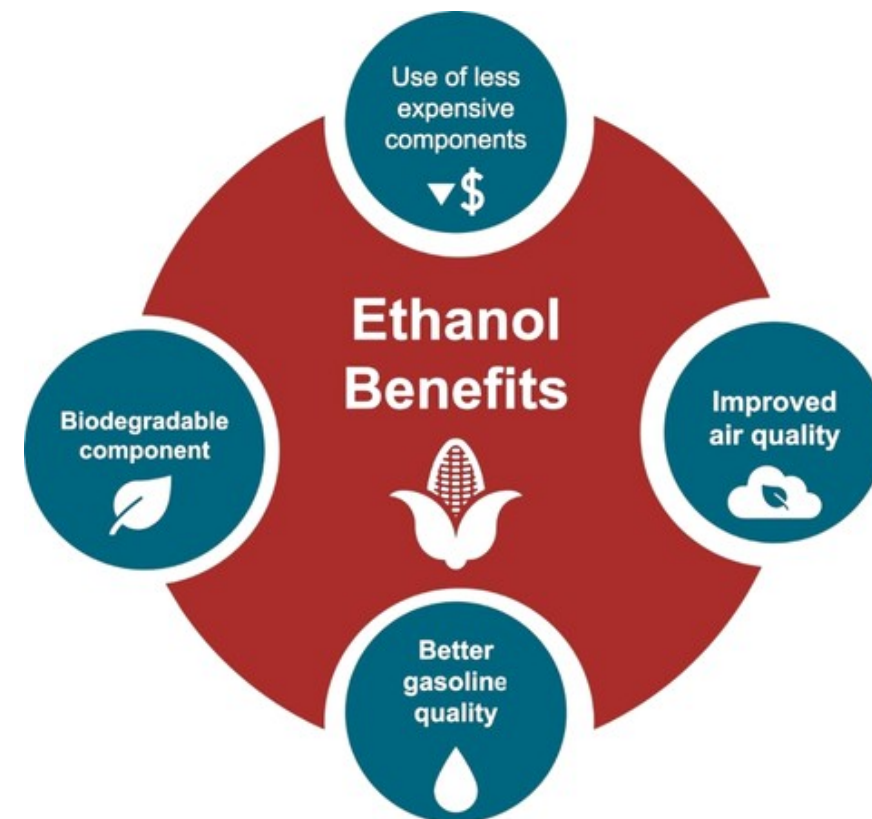
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

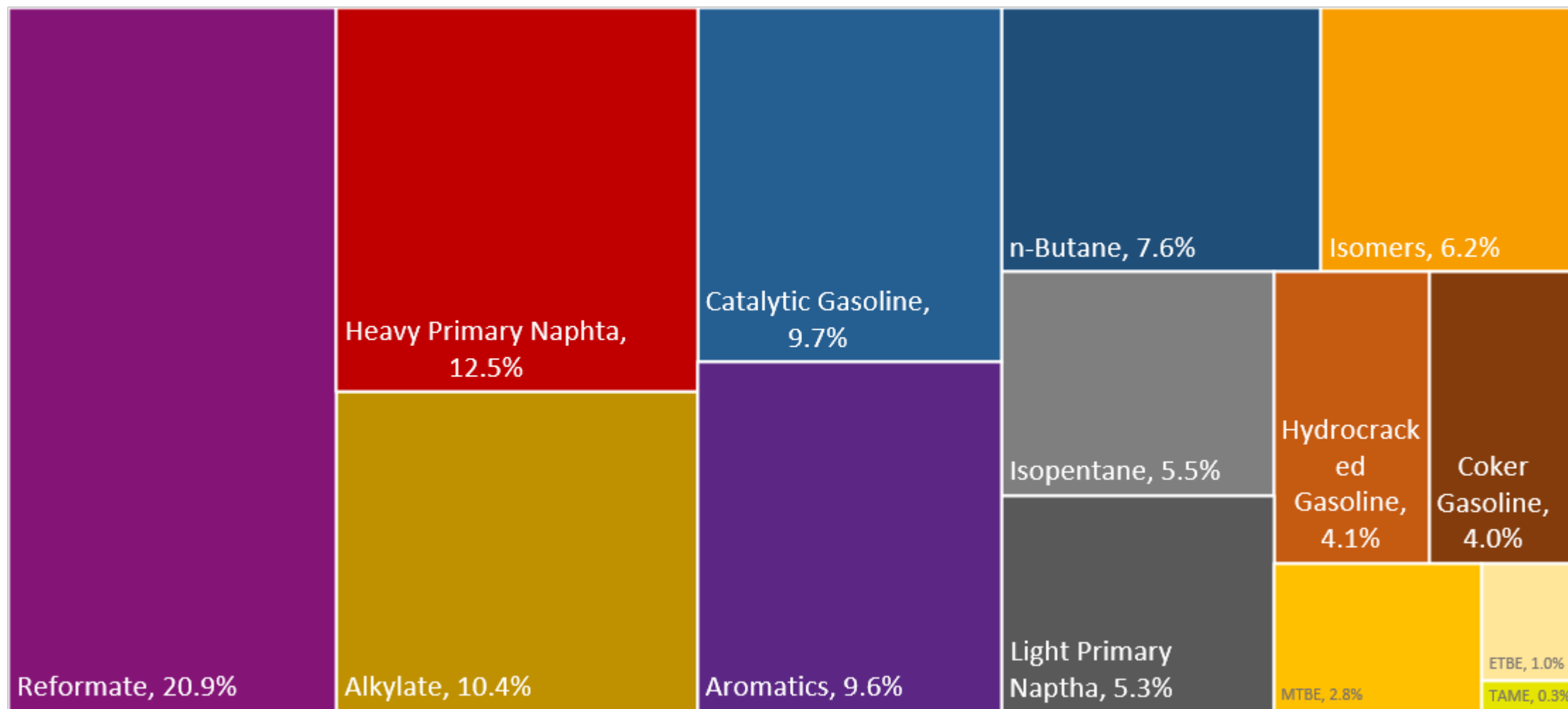
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

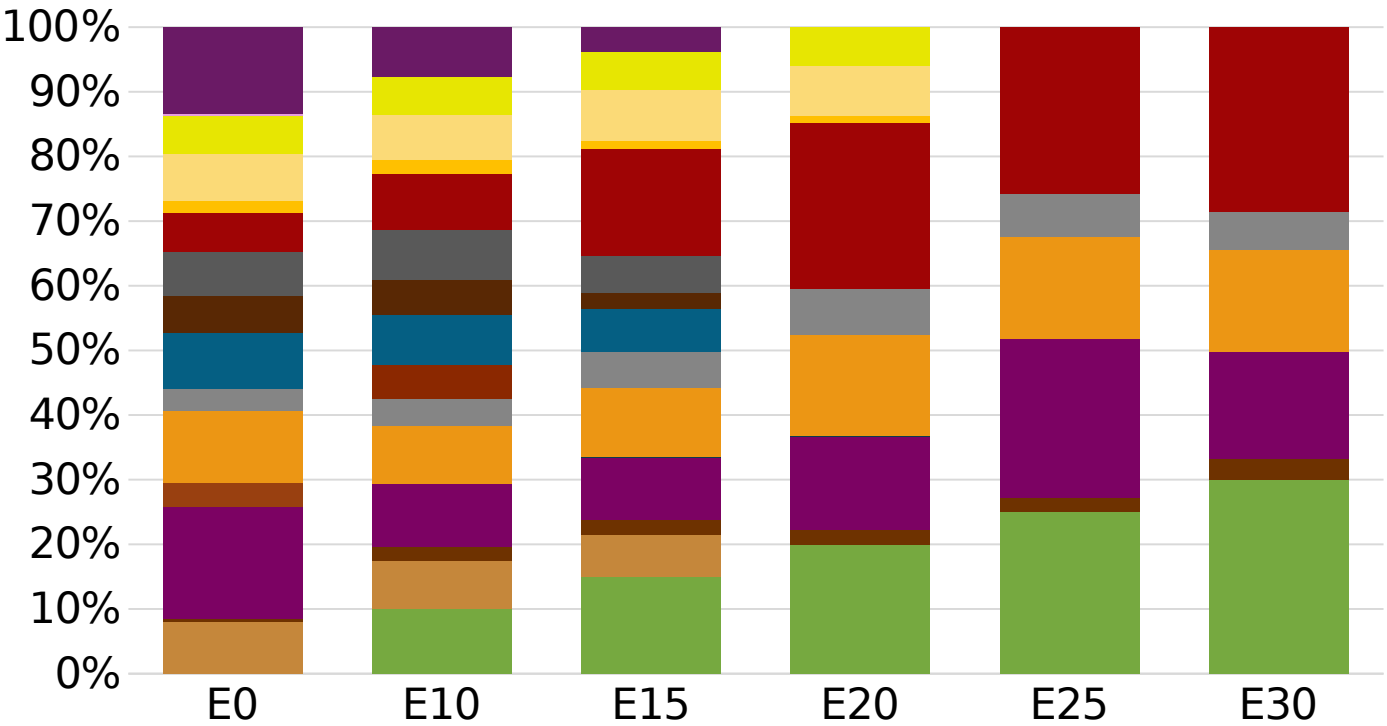
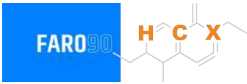
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Finland Available Blending Components



Finland – Gasoline 95 – Constant Octane

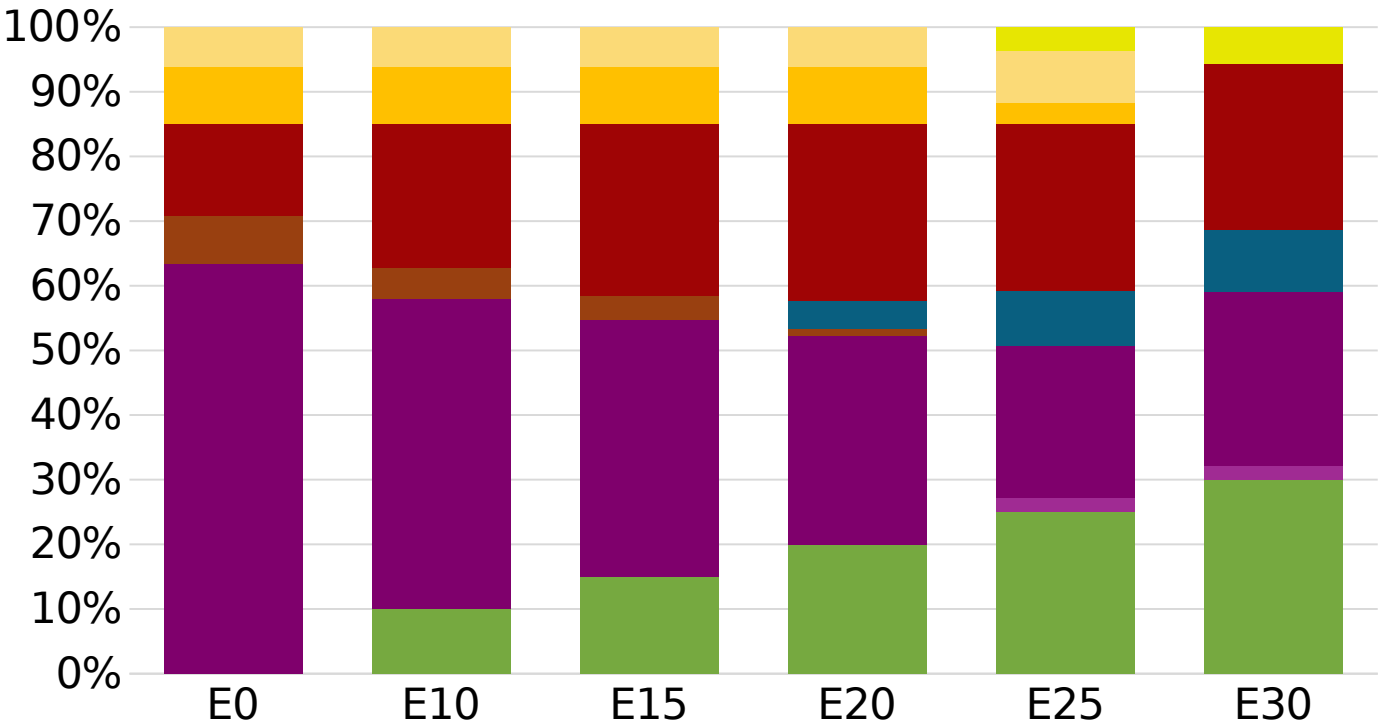
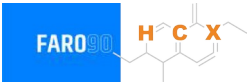


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.091	\$ 1.953	\$ 1.853	\$ 1.738	\$ 1.692	\$ 1.639

Source:
Faro90

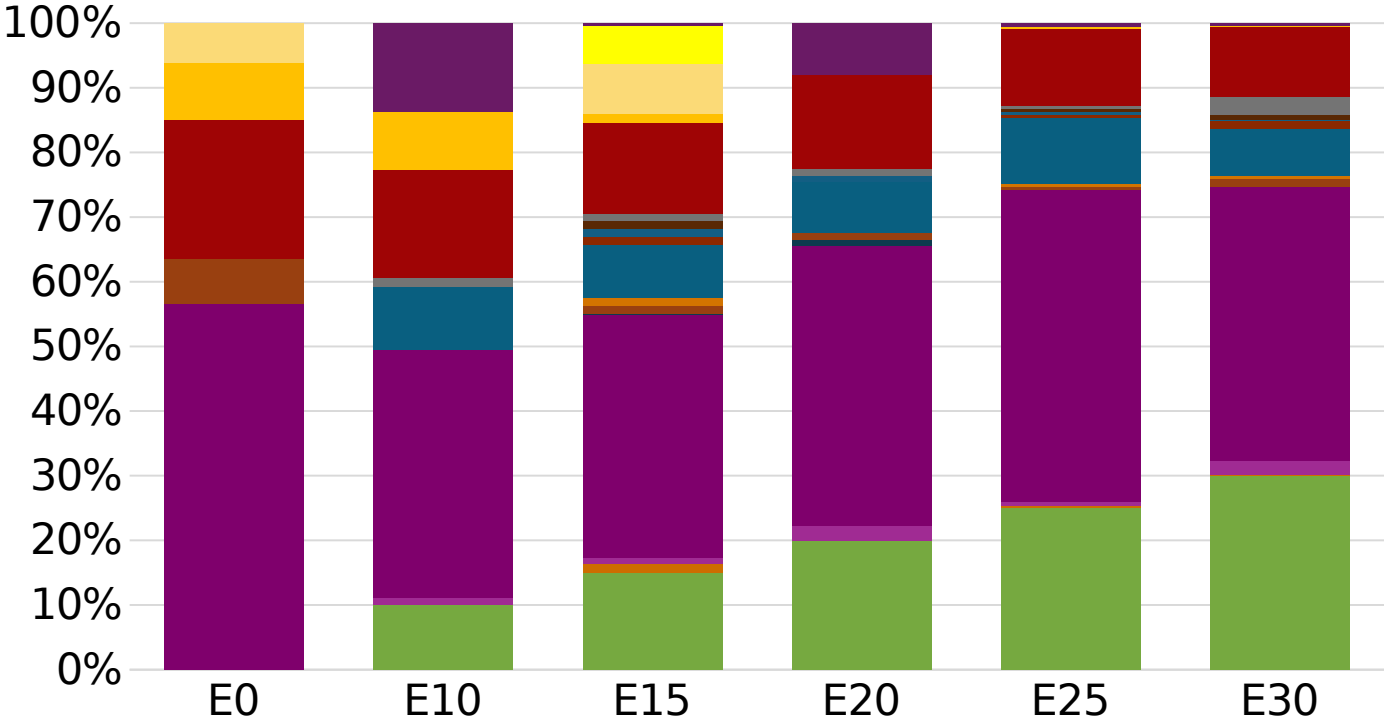
Finland - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.111	\$ 1.994	\$ 1.925	\$ 1.853	\$ 1.788	\$ 1.738

Average CIF 2025
Prices

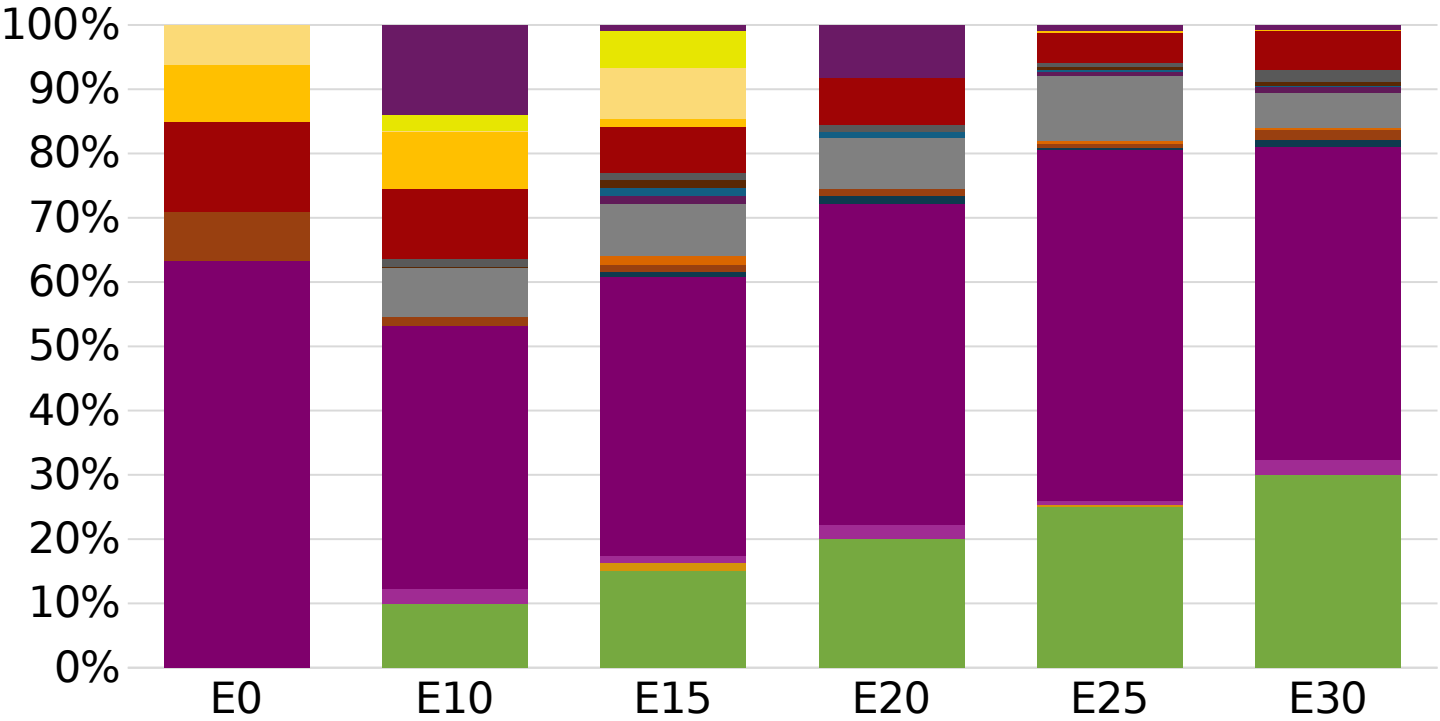
Finland – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979

Finland – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

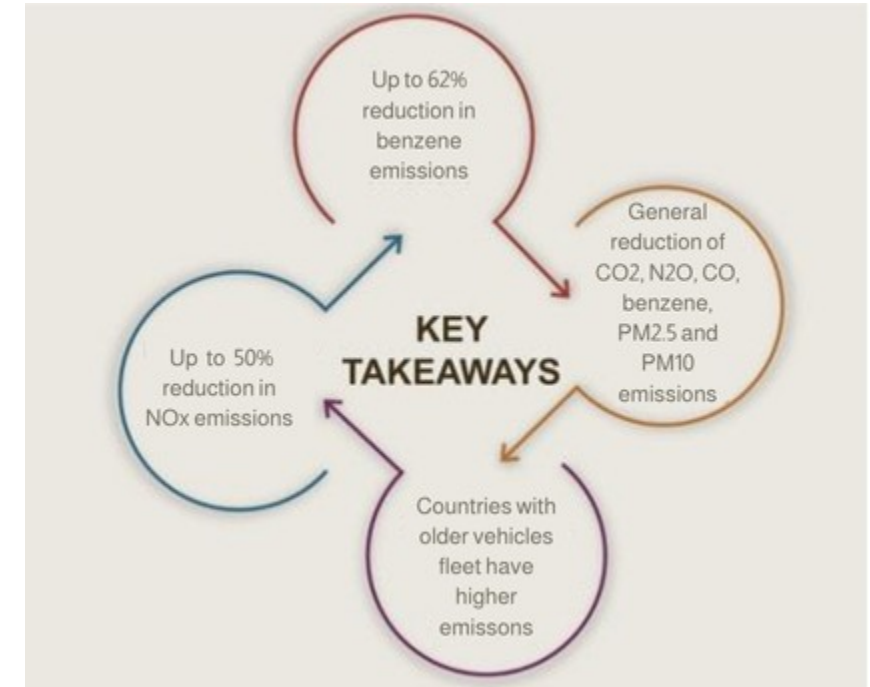
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

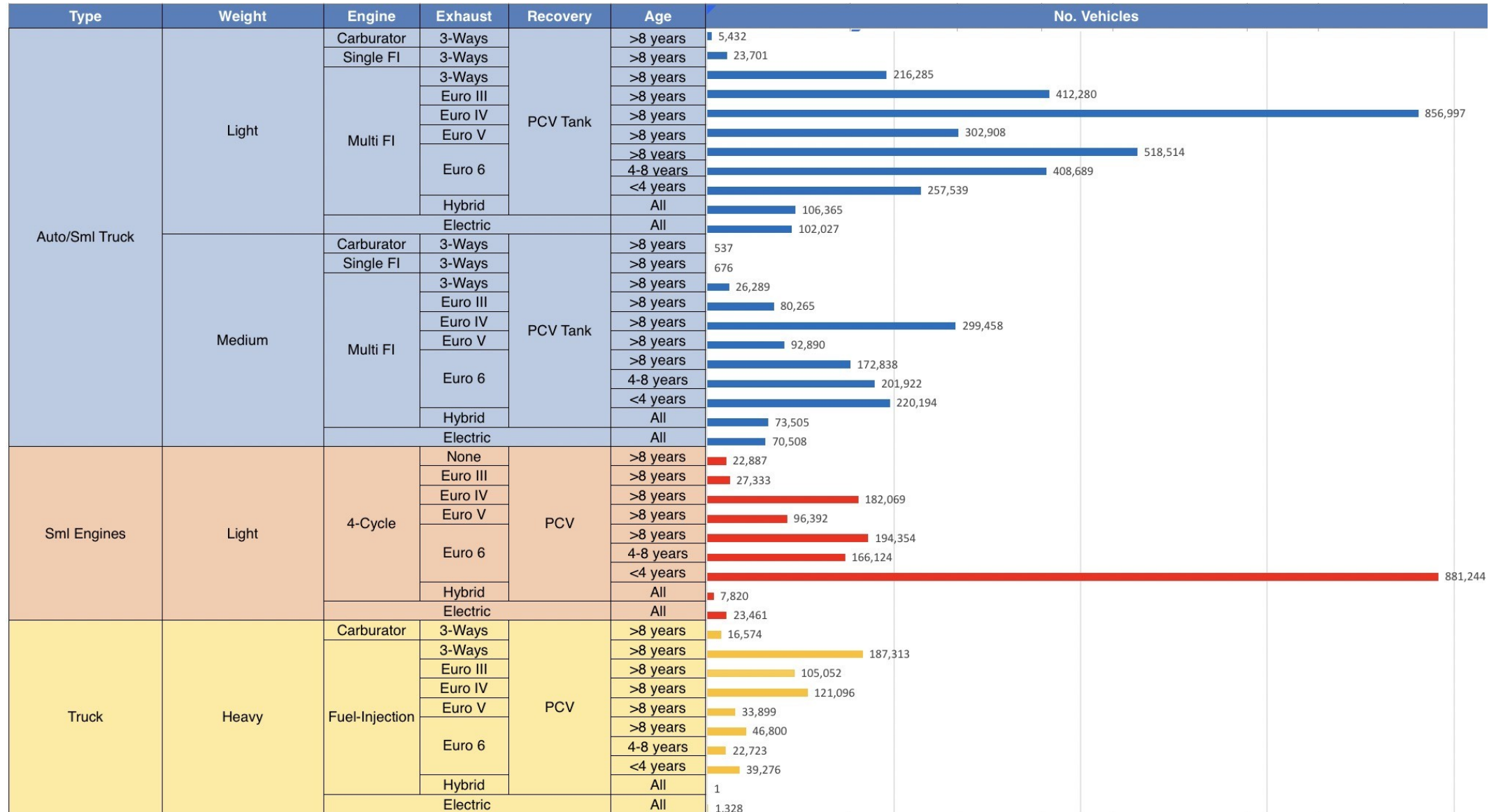
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Finland



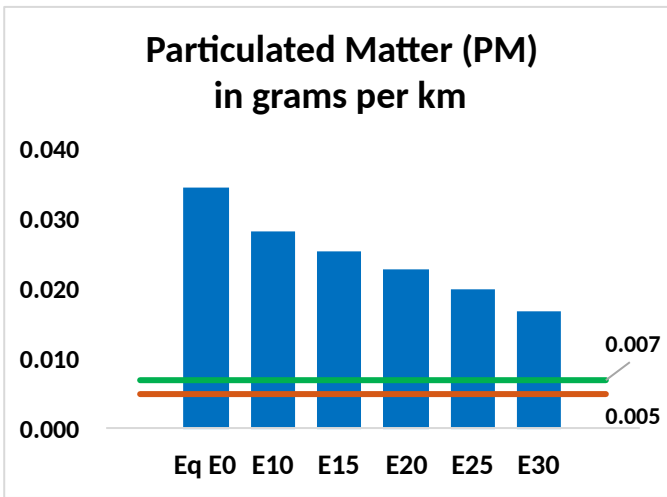
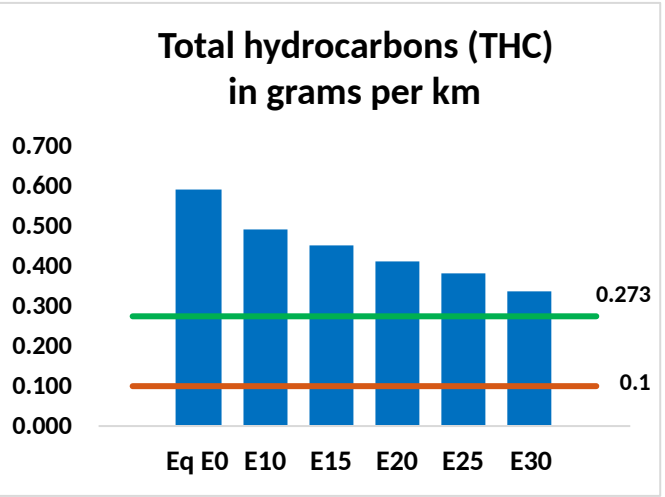
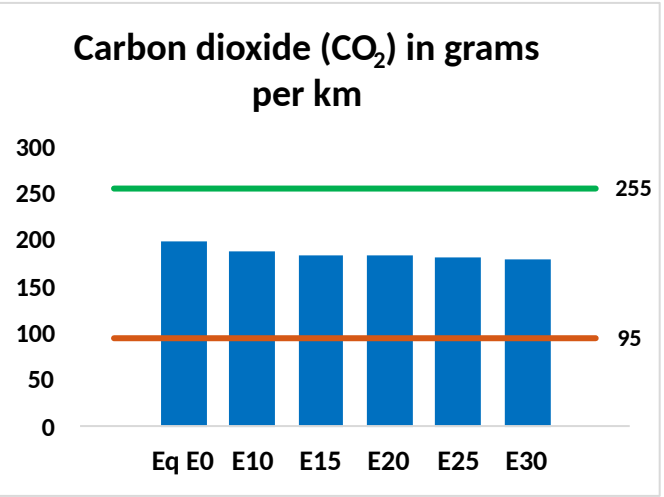
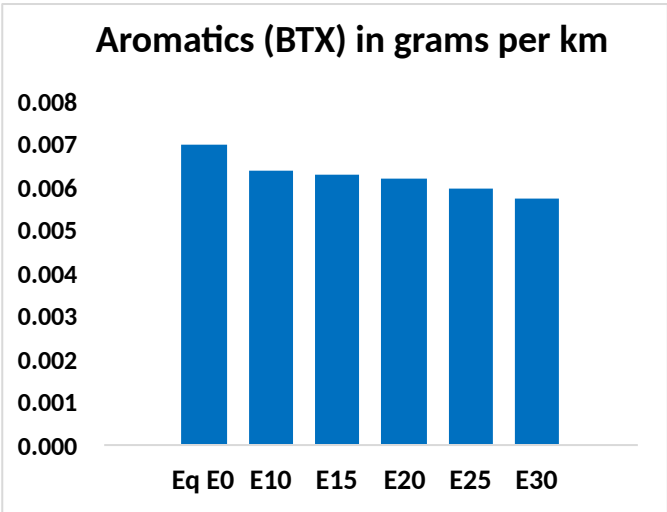
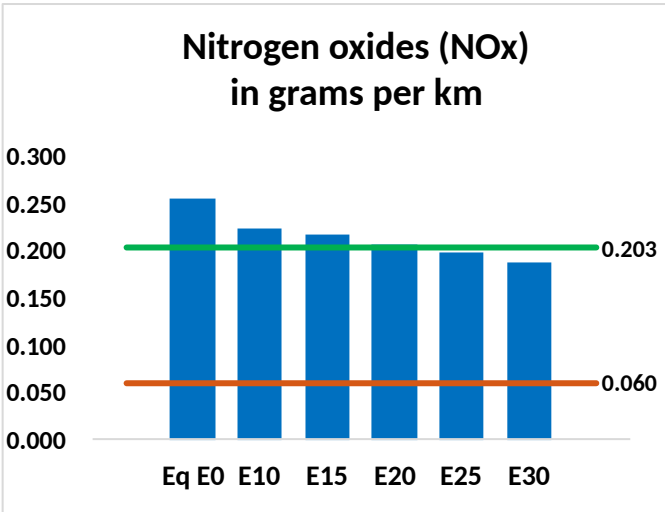
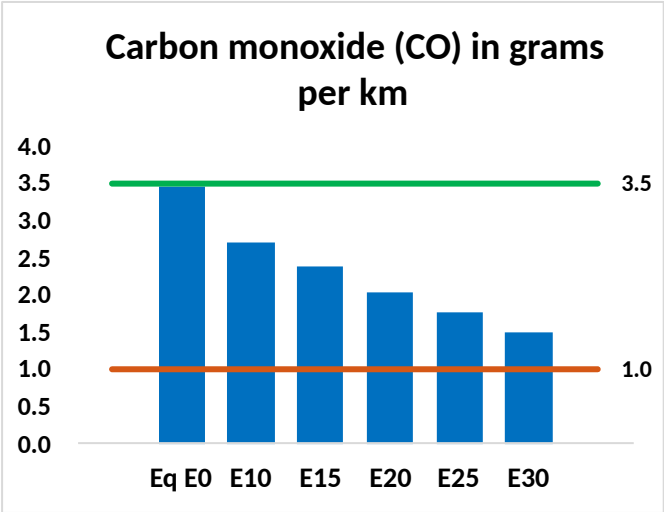
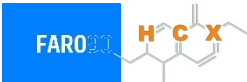
Vehicle Fleet: 6,625,564

Gasoline Vehicle Fleet: 3,473,949

Source: Eurostat, ACEA

Finland – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Finland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	312,839	278,880	244,922	215,160	184,321	160,339	135,854	-22%	-41%
CO2	17,985,650	17,523,162	17,086,367	16,742,841	16,572,851	16,474,128	16,170,462	-5%	-8%
VOC	53,535	49,130	44,726	41,060	37,345	34,481	30,686	-16%	-30%
NOx	23,215	21,435	20,243	19,650	18,679	17,926	17,034	-13%	-20%
SOx	205	189	174	161	148	135	120	-15%	-28%
NH3	6,546	6,481	6,417	6,447	6,520	6,571	6,613	-2%	0%
N2O	276	274	273	275	280	283	287	-1%	2%
CH4	11,529	11,529	11,529	11,702	11,956	12,175	12,382	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	317	300	282	269	257	247	234	-11%	-19%
Acetaldehyde	941	1,080	1,584	2,412	3,287	3,756	4,438	68%	249%
Formaldehyde	3,517	3,419	3,986	4,681	4,904	5,425	5,905	13%	39%
Benzene	636	611	579	570	565	540	523	-9%	-11%
PM2.5	1,075	978	880	794	710	619	525	-18%	-34%
PM10	2,045	1,859	1,673	1,510	1,349	1,178	998	-18%	-34%

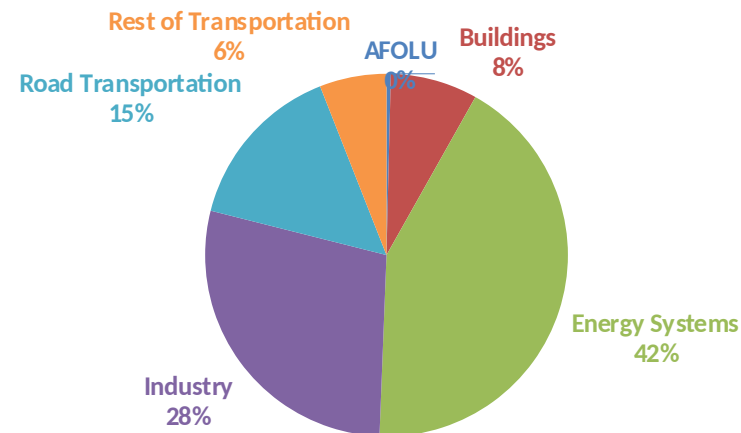
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

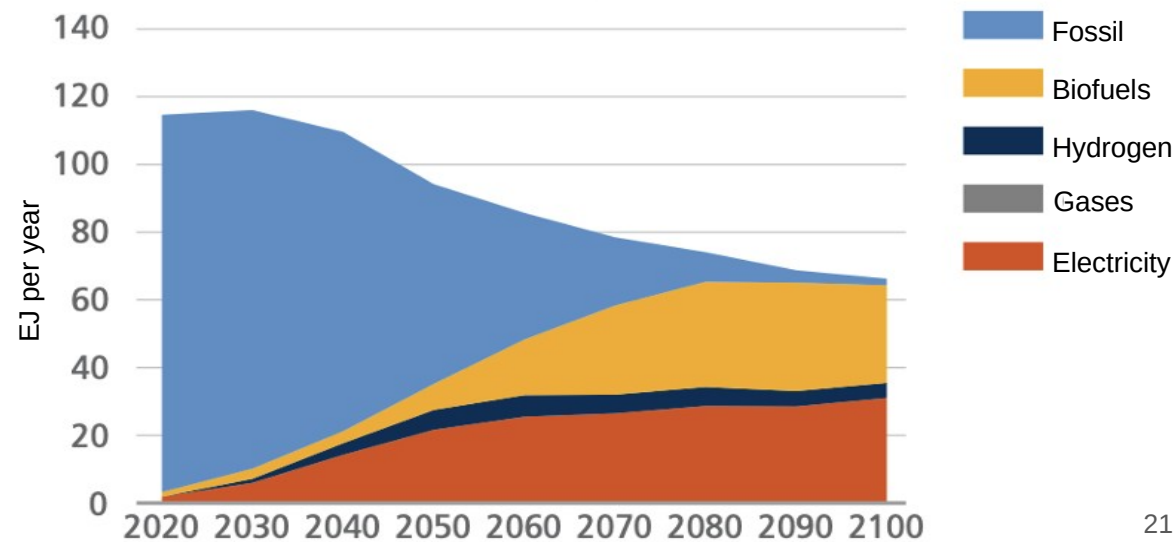
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

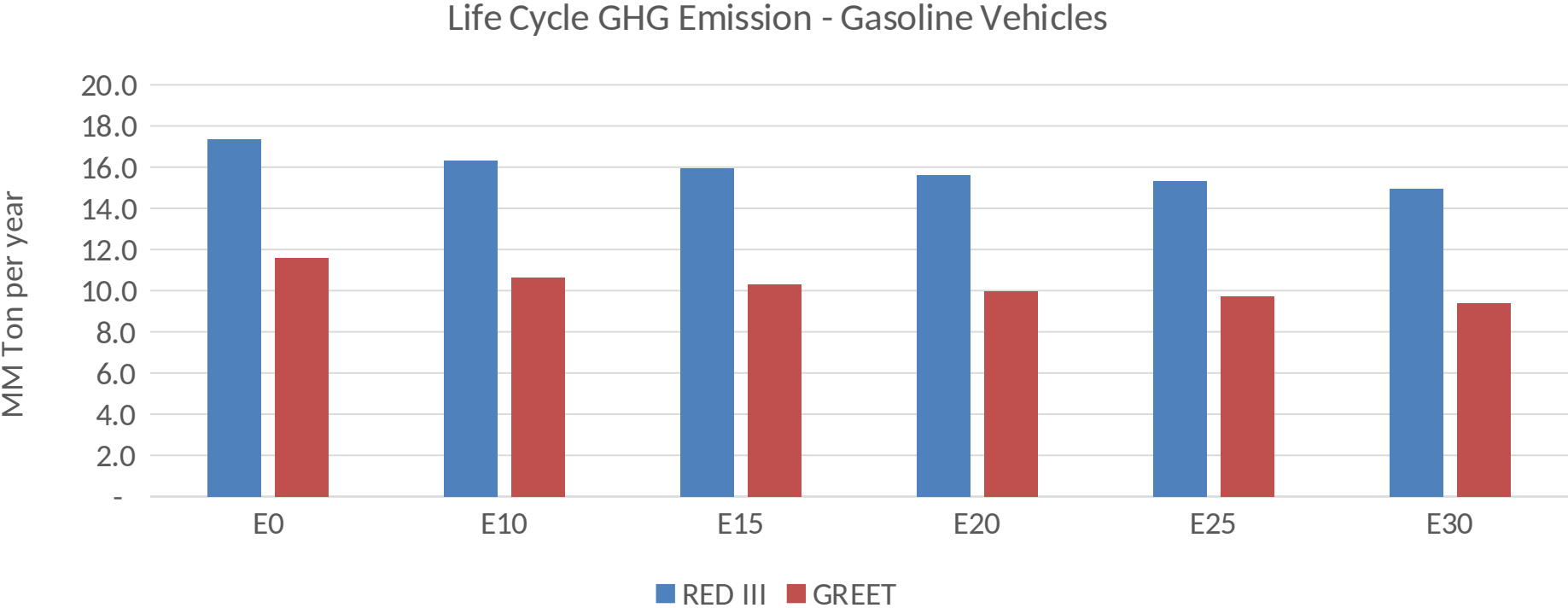
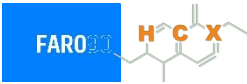
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	64.9
Target @ 2035 (MMT/year)	36.6
Delta	28.4